Machine Learning for Predicting Voiding Dysfunction after Nerve-Sparing Radical Hysterectomy

Experiment Note — Results and Findings

February 2026

1 Objective

To develop and evaluate machine learning models for predicting resumption of spontaneous voiding at postoperative day 7 (POD7) after nerve-sparing radical hysterectomy for early-stage cervical cancer.

Primary outcome: Accuracy of ML prediction models (AUC, accuracy, sensitivity, specificity).
Secondary outcome 1: Factors associated with voiding resumption (SHAP-based feature importance).
Secondary outcome 2: Timing of bladder recovery among patients with POD7 dysfunction.

2 Data and Methods

2.1 Study Population

- **Source:** Single-center retrospective cohort (Chiang Mai University), 2005–2023
- **Total:** 1,087 patients; 1,020 complete cases (93.8%)
- **Outcome:** Voiding dysfunction = post-void residual >100 mL at POD7 (binary)

2.2 Temporal Split

- **Development cohort:** 2005–2014, $n = 701$, events = 329 (46.9%)
- **Validation cohort:** 2015–2023, $n = 319$, events = 111 (34.8%)

Event rate decreased from 46.9% to 34.8% across eras, indicating temporal drift likely reflecting improved surgical technique over time.

2.3 Predictors

26 candidate predictors initially; 25 after dropping pelvic lymph node count (near-zero variance): 8 continuous (age, BMI, parity, preoperative lesion size [wardsize, cm], operative time, average parametrial length, average vaginal cuff length, estimated blood loss) and 17 categorical (disease type, DM, menopause, HIV, prior cesarean, preoperative UTI, conization, prior chemotherapy, stage, incision, radicality class, anesthesia, appendectomy, salpingo-oophorectomy, aortic LN dissection, final histology, tumor appearance).

2.4 Models

Nine models trained with 10-fold stratified cross-validation on the development cohort:

1. Logistic Regression (unpenalized baseline)
2. LASSO (L1)
3. Elastic Net (L1+L2)
4. Random Forest
5. XGBoost
6. Support Vector Machine (RBF kernel)
7. Decision Tree

8. AdaBoost
9. Stacking Ensemble (meta-learner over EN, RF, XGB, SVM, AdaBoost)

All models used GridSearchCV for hyperparameter tuning. Youden's J threshold was derived from development cohort out-of-fold predictions and applied to the validation cohort. Because the optimal threshold depends on prevalence, this threshold may not be optimal for the validation cohort (event rate 34.8% vs. 46.9% in development); AUC (threshold-independent) is therefore the primary performance metric, and threshold-dependent metrics (sensitivity, specificity) should be interpreted cautiously. Development AUC estimates were obtained from cross-validated out-of-fold predictions of the tuned models, which may introduce mild optimistic bias (non-nested CV). The temporally separated validation cohort provides a fully independent performance assessment unaffected by this bias; the empirical observation that development and validation AUCs agree closely (Table 1) is consistent with negligible optimism.

2.5 Two-Stage Approach

- **Stage 1:** All 25 predictors
- **Stage 2:** SHAP-based consensus feature selection (top 14 predictors), models refit

Feature importance was assessed using SHAP TreeExplainer (XGBoost, Random Forest), absolute coefficients (Elastic Net), and permutation importance (SVM). Rankings were aggregated by mean rank across methods to avoid scale-dependency (Elastic Net coefficients reflect standardized continuous variables but raw-scale binary dummies). Note: feature importance was computed on the full development set prior to cross-validation refit, which may introduce selection bias in development CV estimates; however, the validation cohort was held out entirely during feature selection, so validation AUC estimates remain unbiased. Consensus across four independent methods further mitigates this concern.

3 Results

3.1 Stage 1: All Models, All 25 Predictors

Table 1: Model performance. Dev AUC from 10-fold CV out-of-fold predictions; Val metrics on temporal validation ($n = 319$). Thresholds derived from development cohort. Sorted by Val AUC.

Model	Dev AUC	Val AUC (95% CI)	Acc	Sens	Spec	Brier
Stacking	0.688	0.695 (0.636–0.757)	0.674	0.523	0.755	0.236
RCS Logistic†	0.685	0.691 (0.627–0.752)	0.715	0.369	0.899	0.206
SVM	0.689	0.691 (0.631–0.752)	0.683	0.387	0.841	0.211
AdaBoost	0.668	0.690 (0.630–0.752)	0.683	0.324	0.875	0.219
Elastic Net	0.683	0.684 (0.626–0.745)	0.655	0.577	0.697	0.216
Random Forest	0.675	0.683 (0.618–0.747)	0.611	0.649	0.591	0.221
XGBoost	0.685	0.677 (0.615–0.739)	0.668	0.288	0.870	0.221
LASSO	0.682	0.676 (0.619–0.736)	0.652	0.468	0.750	0.218
Decision Tree	0.649	0.646 (0.580–0.709)	0.483	0.802	0.312	0.230
Logistic	0.667	0.631 (0.572–0.695)	0.624	0.477	0.702	0.240

†RCS Logistic = restricted cubic splines on age and parity (3 knots each) + bmi + wardsize + ave_vgcufflength. Uses only 5 predictors; replicates the R analysis from the clinical team. Included as a standalone comparator (not part of the 9-model ML pipeline) to benchmark ML against classical regression with domain-selected variables.

All models clustered between AUC 0.63–0.70. Using the pre-specified reference model (Elastic Net, pre-specified as the reference based on its calibration profile rather than post-hoc by highest AUC), no comparison reached statistical significance after Bonferroni correction (8 tests, each model vs. the reference). The test is a paired bootstrap ($B = 2000$): on each bootstrap resample of the same $n = 319$ validation patients, the AUC difference (model − reference) is computed; the two-sided p-value is the proportion of resamples where the observed difference flips sign. Only Logistic was significant before correction ($p_{\text{raw}} = 0.002$, $p_{\text{Bonf}} = 0.016$). This confirms that all top models achieve comparable discrimination.

3.2 Calibration

While AUC measures a model's ability to *rank* patients (discrimination), **calibration** measures whether the predicted probabilities are *accurate* — i.e., among patients predicted to have 30% risk, do approximately 30% actually develop dysfunction? Three metrics are used: **CITL** (calibration-in-the-large) = $\text{logit}(\bar{y}) - \overline{\text{logit}(\hat{p})}$; ideal = 0. Negative values indicate the model's average predicted log-odds exceeds the observed log-odds (systematic over-prediction on the probability scale). **Calibration slope** measures how well the model spreads risk across patients; ideal = 1.0. In the calibration model $\text{logit}(P) = a + b \cdot \text{logit}(\hat{p})$: slope $b < 1$ indicates overfitting (predictions too extreme, need shrinkage); slope $b \gg 1$ indicates underfitting (predictions compressed into a narrow range — the model fails to differentiate patients, as with Stacking's slope of 10.7). **Brier score** combines both discrimination and calibration into a single metric (lower = better, range 0–1).

Table 2: Calibration metrics on validation cohort ($n = 319$). CITL = calibration-in-the-large; ideal slope = 1.0. Bootstrap 95% CIs ($B = 2000$) for original slope. Recalibrated CITL ≈ 0 is expected by construction (the intercept update is specifically designed to match the observed event rate).[a]

	Original			Recalibrated		
Model	**CITL**	**Slope (95% CI)**	**Brier**	**CITL**	**Slope**	**Brier**
RCS Logistic	−0.214	0.945 (0.63–1.34)	0.206	—	—	—
Elastic Net	−0.396	0.978 (0.67–1.37)	0.216	0.079	0.958	0.206
SVM	−0.368	1.119 (0.78–1.55)	0.211	0.067	1.095	0.202
LASSO	−0.387	0.900 (0.61–1.27)	0.218	0.088	0.882	0.208
AdaBoost	−0.452	1.767 (1.18–2.48)	0.219	0.025	1.754	0.207
Random Forest	−0.470	1.951 (1.27–2.80)	0.221	0.023	1.923	0.209
XGBoost	−0.423	2.049 (1.30–2.89)	0.221	0.016	2.011	0.212
Stacking	−0.495	10.654 (7.39–14.65)	0.236	0.001	10.170	0.222

[a] Brier score 95% CIs: RCS Logistic 0.189–0.224; Elastic Net 0.200–0.232; SVM 0.196–0.226; LASSO 0.203–0.234; AdaBoost 0.208–0.229; Random Forest 0.212–0.231; XGBoost 0.213–0.231; Stacking 0.232–0.239. Decision Tree (slope 0.53) and unpenalized Logistic (slope 0.21) are omitted due to poor discrimination (Table 1); calibration metrics are not clinically meaningful for models with inadequate discrimination. RCS Logistic was not recalibrated as it serves as the clinical team's reference model.

Elastic Net has the calibration slope point estimate closest to 1.0 among full models (0.978, 95% CI 0.67–1.37), though the wide CI overlaps substantially with LASSO (0.900; 0.61–1.27) and SVM (1.119; 0.78–1.55), so the ranking is uncertain given the limited validation sample ($n = 319$). EN is recommended for clinical deployment based on the combination of competitive AUC and favorable calibration point estimates.

Recalibration-in-the-large adjusts for systematic over- or under-prediction by shifting the model's intercept so that the mean predicted probability matches the observed event rate in a new population. This is a standard technique when models are applied to populations with different baseline risk (e.g., temporal drift). The method constrains the calibration slope to 1 (offset-only update), correcting the overall level of predictions without altering how the model spreads risk across patients. CITL approaching zero after recalibration is expected by construction — the intercept update is specifically designed to achieve this. The more informative question is whether the calibration *slope* improves, which offset-only recalibration does not address.

Negative CITL across all original models reflects temporal drift: the event rate dropped from 46.9% (dev) to 34.8% (val), causing systematic over-prediction. Recalibration corrected this for all models, though residual CITL of 0.07–0.09 for EN and LASSO reflects numerical convergence tolerance in the offset optimization (Brent's method). Recalibrated slopes differ minimally from originals ($\Delta < 0.04$ for all except Stacking) because the offset is a constant shift on the log-odds scale; the Stacking delta (0.484) reflects numerical sensitivity of extreme-slope estimates. Tree-based models (Random Forest slope 1.92, XGBoost 2.01, AdaBoost 1.75) continue to show compressed predictions requiring further calibration beyond simple intercept correction. Stacking's extreme slope (≈ 10.7) indicates near-constant probability output from the meta-learner, making it unsuitable for clinical probability estimation. Note: recalibration was estimated on the same validation set used for evaluation; the recalibrated metrics therefore represent *apparent* validation performance and would require assessment on a third independent dataset for true external validation of

the recalibrated model.

Note on Stacking: Stacking is **unsuitable for clinical deployment** despite achieving the highest AUC (0.695). Its calibration slope of 10.7 (95% CI 7.39–14.65) indicates that the meta-learner outputs near-constant probabilities — the model cannot meaningfully distinguish between a 30% and a 60% risk patient. While offset-only recalibration corrects the overall mean (CITL → 0.001), it cannot fix the slope; the model still lacks individual-level probability differentiation. Stacking is retained in all tables for transparency only. **The recommended model is Elastic Net**, which combines competitive AUC (0.684) with the calibration slope closest to 1.0 among full models (0.978, 95% CI 0.67–1.37). This case illustrates that **discrimination (AUC) alone is insufficient** for evaluating clinical prediction models — calibration must be jointly assessed.

3.3 Feature Importance — Consensus Ranking

Table 3: Consensus feature ranking across 4 methods (XGBoost SHAP, RF SHAP, EN |coefficients|, SVM permutation). EN ranks for categorical variables (e.g., stage, incision) may be inflated due to scale differences between standardized continuous features and raw binary dummies; the rank-based consensus mitigates but does not eliminate this bias.

Variable	XGB	RF	EN	SVM	Mean Rank
Ave vaginal cuff length	1	1	1	1	**1.00**
BMI	3	3	7	2	3.75
Radicality class	2	4	2	13	5.25
Preop lesion size	5	2	11	4	5.50
Age	4	7	10	3	6.00
Ave parametrial length	7	6	12	7	8.00
Parity	6	12	15	5	9.50
Stage	18	8	3	9	9.50
Incision	21	11	4	6	10.50
Salpingo-oophorectomy	12	10	6	15	10.75

Average vaginal cuff length is the dominant predictor — ranked #1 by all four methods. This is a proxy for radicality of dissection near the bladder innervation. BMI, radicality class, preoperative lesion size (wardsize), and age round out the top 5.

SHAP interpretation of top features (Figure 4):

- **Ave vaginal cuff length:** The strongest and most consistent predictor. Higher values (longer resection) push SHAP strongly positive (toward dysfunction), while shorter cuff lengths are protective. This is clinically expected — more extensive paravaginal dissection increases the risk of pelvic autonomic nerve injury. The SHAP dependence plot (Figure 6) shows a roughly monotonic positive relationship, suggesting this effect is approximately linear.
- **Radicality class** (one-hot: class 3): Patients *not* in radicality class 3 (i.e., less radical surgery) show strongly negative SHAP values, indicating lower risk. Class 3 (the most radical classification commonly performed) has near-zero or positive SHAP. This aligns with the known dose-response relationship between surgical radicality and nerve damage.
- **BMI:** Higher BMI pushes SHAP moderately positive (increased risk). Possible mechanisms include more technically difficult dissection in obese patients and slower postoperative autonomic recovery. The effect is moderate and approximately linear.
- **Age:** Older patients show positive SHAP values, consistent with age-related decline in peripheral nerve recovery capacity. The SHAP dependence plot suggests a gradual increase rather than a sharp age threshold.
- **Preoperative lesion size (wardsize):** Size measured at preoperative evaluation (1 day preop, cm). Moderate importance with mixed directionality. Larger preoperative lesion size may necessitate more radical dissection, indirectly increasing nerve injury risk.

Non-Linear Relationships

A key motivation for applying ML was the potential to capture non-linear predictor-outcome relationships that logistic regression might miss. Visual inspection of the SHAP dependence plots (Figure 6) suggests that the top predictors exhibit **predominantly linear or weakly non-linear** relationships with the outcome. Average vaginal cuff length and age show monotonic positive trends; BMI is approximately linear. No strong threshold effects, U-shaped curves, or complex interactions are apparent.

This is consistent with our finding that RCS logistic regression (which captures mild non-linearity via 3-knot cubic splines) matches the best ML models in discrimination (AUC 0.691 vs. 0.695). The ML models did not uncover clinically meaningful non-linear patterns beyond what restricted cubic splines already capture. The AUC ceiling of ∼0.70 is therefore not due to modeling limitations but rather to the inherent predictive information available in the measured variables.

3.4 Feature-Selected Models

We evaluated two reduced predictor sets to test how much signal concentrates in fewer variables: (a) a **5-variable model** selected by the clinical team via domain knowledge and stepwise logistic regression in R, and (b) a **14-variable model** selected by data-driven SHAP consensus ranking.

3.4.1 5-Variable Model (Clinical Selection)

The clinical team's R analysis selected 5 predictors using domain expertise and backward stepwise regression on the full dataset ($n = 1{,}020$): **age, BMI, parity, preoperative lesion size (wardsize), and average vaginal cuff length**. Note: the stepwise selection used all available data (development + validation); when repeated on the development cohort alone, backward stepwise selected `ave_pmlength` in place of `parity`, suggesting moderate instability in variable selection. The selected 5 overlap substantially with the ML-derived top 5 from Table 3 (4 of 5 match; the exception is parity vs. radicality class at rank #3). This convergence between clinical judgment and data-driven selection validates both approaches.

The R analysis used restricted cubic splines (RCS): `rcs(age,3) + bmi + rcs(parity,3) + wardsize + ave_vgcufflength`. We replicated this in Python to cross-validate.

Table 4: Python replication of R's RCS logistic model. Knots at 10th/50th/90th percentiles of development cohort.

Metric	R result	Python result
Apparent AUC (dev, resubstitution)	0.6988	0.6950
10-fold CV AUC (dev)	—	0.685 (0.645–0.723)
Temporal validation AUC	reported	0.691 (0.627–0.752)
Val Brier	—	0.206
Val CITL / Slope	—	−0.214 / 0.945

The R analysis is reproducible in Python (apparent AUC 0.6950 vs. 0.6988; minor knot placement differences). We then ran all 9 ML models on the same 5 variables to test whether ML could extract more signal:

Table 5: All 10 models restricted to 5 clinically-selected variables. Sorted by Val AUC.

Model	Val AUC (95% CI)	Brier	Acc	Sens	Spec	Slope
RCS Logistic	0.691 (0.627–0.752)	0.206	0.715	0.369	0.899	0.945
Elastic Net	0.690 (0.629–0.754)	0.208	0.715	0.477	0.841	1.075
Logistic	0.690 (0.629–0.753)	0.207	0.718	0.468	0.851	0.984
LASSO	0.690 (0.628–0.753)	0.207	0.718	0.468	0.851	0.989
Stacking	0.685 (0.620–0.751)	0.234	0.705	0.387	0.875	8.756
SVM	0.683 (0.621–0.747)	0.209	0.693	0.405	0.846	0.982
AdaBoost	0.676 (0.613–0.740)	0.219	0.687	0.450	0.812	1.471
Random Forest	0.673 (0.608–0.738)	0.216	0.687	0.324	0.880	1.366
XGBoost	0.663 (0.594–0.729)	0.222	0.687	0.297	0.894	1.851
Decision Tree	0.639 (0.575–0.701)	0.225	0.602	0.541	0.635	0.684

5-variable Elastic Net calibration: CITL = −0.309, slope = 1.075 (95% CI 0.71–1.51), Brier = 0.208 (0.192–0.224). The 5-var CITL (−0.309) is worse than RCS Logistic (−0.214), reflecting slightly more over-prediction; both indicate temporal drift from the higher development event rate. RCS Logistic calibration: CITL = −0.214, slope = 0.945 (0.63–1.34), Brier = 0.206 (0.189–0.224).

Result: With only 5 variables, **no ML model outperforms the RCS logistic**. All linear models converge to AUC ≈ 0.690, matching RCS Logistic (0.691). Complex models (XGBoost, Stacking) perform *worse*, likely overfitting with so few features. The 5-variable models achieve slightly higher AUC than the 25-variable versions (e.g., 5-var EN 0.690 vs. 25-var EN 0.684), consistent with removal of noisy features that dilute signal; this difference is within sampling noise. The 5-variable Elastic Net (slope 1.075) shows slight underfitting — predictions are marginally compressed toward the mean, meaning high-risk patients may be slightly underestimated — compared to the 25-variable version's slight overfitting (slope 0.978). Both are clinically acceptable deviations from ideal.

3.4.2 14-Variable Model (SHAP Consensus Selection)

The 14 predictors were selected from the consensus ranking (Table 3) by taking the top half of features by mean rank (mean rank ≤ median). This yielded 14 features with mean ranks ≤ 12.25, while the remaining 11 had mean ranks ≥ 13.25. The selected 14 are: ave_vgcufflength, BMI, radicality class, preoperative lesion size (wardsize), age, ave_pmlength, parity, stage, incision, salpingo-oophorectomy, EBL, tumor appearance, prior chemotherapy, and operative time. The 11 excluded features (preop UTI, anesthesia, disease type, conization, prior cesarean, DM, menopause, final histology, appendectomy, HIV, aortic LN) had weak and inconsistent importance across methods. Note: feature importance was computed on the full development set before cross-validation refit (see Limitations); the cutoff is a pragmatic choice rather than a statistically derived threshold.

Table 6: Stage 1 (25 vars) vs. Stage 2 (14 vars) — top models on validation. AUC point estimates without CIs; all differences are within the sampling noise of Table 1 CIs. Note: Stage 2 development AUCs (not shown) may carry mild upward bias from feature selection on the full development set; validation AUCs are unaffected.

Model	Stage	Val AUC	Brier	Acc	Sens	Spec
S2 Random Forest	S2 (14)	0.699	0.215	0.677	0.468	0.788
S2 Stacking	S2 (14)	0.695	0.219	0.680	0.495	0.779
S1 Stacking[‡]	S1 (25)	0.695	0.236	0.671	0.063	0.995
S1 SVM	S1 (25)	0.691	0.211	0.674	0.486	0.774
S2 SVM	S2 (14)	0.690	0.212	0.683	0.532	0.764
S2 Elastic Net	S2 (14)	0.689	0.215	0.683	0.550	0.755
S1 Elastic Net	S1 (25)	0.684	0.216	0.668	0.541	0.736

[‡]S1 Stacking sensitivity 6.3% with specificity 99.5% indicates the model predicts nearly all patients as negative at the development-derived Youden threshold — a degenerate operating point consistent with its poor calibration (slope 10.7).

Feature reduction from 25 to 14 predictors **maintained** validation AUC across all model types (differences within statistical noise; no formal test reached significance given overlapping confidence intervals). Stage 2 models showed comparable or marginally lower Brier scores, suggesting the removed 11 features contributed noise rather than signal.

3.4.3 Summary: Signal Concentration

The progression 25 → 14 → 5 predictors with stable AUC (∼0.69–0.70) demonstrates that predictive signal is concentrated in a small number of variables. For the recommended Elastic Net specifically: 25-var Val AUC 0.684, 14-var 0.689, 5-var 0.690 — a consistent pattern of maintained or marginally improved performance with fewer predictors. This has important implications:

1. The AUC ceiling of ∼0.70 is a property of the data, not the modeling approach.
2. Domain-knowledge-driven variable selection with splines matches or beats complex ML.
3. ML adds no incremental discrimination over standard logistic regression for these 5 predictors.
4. The additional 20 predictors in the full models contribute negligible signal.

Attempting to Break the Ceiling: Interaction and Polynomial Features

We tested whether adding polynomial (degree 2) and pairwise interaction terms for the top 5 continuous variables could push beyond AUC ∼0.70 using Elastic Net. Feature counts below include all 25 original predictors (41 columns after one-hot encoding) plus the new polynomial/interaction terms derived from the 5 continuous variables:

Approach	**Features**	**CV AUC (Dev)**	**Val AUC**	**Val 95% CI**
Baseline	41	0.683	**0.684**	0.626–0.745
Poly degree-2	77	0.691	0.660	0.596–0.725
Interactions	51	0.689	0.651	0.589–0.714
Poly + Inter.	62	0.681	0.678	0.618–0.739

All three augmented approaches **decreased** validation AUC despite modest CV improvements — a classic overfitting pattern. The regularizer compensated by selecting much higher penalty (C = 0.01–0.1), effectively shrinking the added features to zero. This confirms the AUC ceiling is a **data-level limitation** (insufficient signal in available predictors), not a feature-engineering or modeling limitation.

Why the ceiling exists: The strongest individual feature-to-outcome correlation is only $r = 0.26$ (ave_vgcufflength with pod7voiding), with most features showing $|r| < 0.10$. The inter-feature correlations (Figure 8) are also modest (all $|r| < 0.40$), limiting multicollinearity but also limiting the scope for interaction-based improvement. The dominant unmeasured confounder is *surgeon technique*: Nantasupha 2018 reported inter-surgeon dysfunction rates ranging from 21% to 72%. This individual variation accounts for more outcome variance than all measured preoperative predictors combined, and cannot be captured by any model using only pre- and intra-operative variables.

4 Comparison with Published Studies

Table 7: Published POUR prediction models. Studies grouped by surgical context. Dev = development, Val = validation, NR = not reported. Studies sorted by relevance to our context.

Study	**Surgery**	n **(dev/val)**	**Dev AUC**	**Val AUC**	**ML?**	**Ext Val?**	**Notes**
Gynecologic / Radical Hysterectomy							
Nantasupha 2018	NS-RH	755/—	NR	NR	No	No	Same institution. Risk factors only (tumor size, UTI, surgeon)
Zuo 2024	RH + PLND	1101/205	0.897	NR[a]	No	Yes	BMJ Open. Prospective ext val. Predictors include postop variables (UTI, pain)
Meng 2024	Type C2 RH	438 (split)	0.819	0.795	No	No	Laparoscopic vs. open comparison. POUR 38–54%
Shi 2025	Gyn abd-pelvic	1264/436	**0.997**	0.676	Yes	Yes	12 ML models. GBDT best. **Massive overfit** (0.997→0.676)
Kim 2024	Prolapse	1122 (split)	0.73	0.73	No	No	Nomogram. Preop PVR, concomitant procedures
Spine Surgery							
Porche 2021	Lumbar	578/235	0.808	0.737	Yes	No	LR + MLP neural net. Internal split only
Porche 2023	Lumbar	—/171	—	0.663	Yes	Yes	Prospective val of above. Sensitivity collapsed (68%→23%)
Tian 2023	Lumbar fusion	294/129	0.891[b]	0.854	Yes	No	RF dev AUC = **0.996** (overfit). Small sample
Kwas 2025	Lumbar decomp	1938 (split)	0.725	0.709	No	No	JBJS. Nomogram. Male, age>65, BPH history
Other Surgical Contexts							
Wei 2021	Thoracic	1484/646	0.77	0.72	No	Yes	Best-validated non-gyn study. Temporal validation
Axelsen 2006	RH	∼396/—	NR	NR	No	No	Survey. Long-term retention, 78% not bothered
Ketelaers 2021	Hernia repair	244/—	0.81	—	No	No	POUR score. CHF (OR 9.5), DM. No validation
Liu 2024	Anorectal	609/—	0.693	0.680[c]	No	No	Nomogram. Male+BPH, pain, epidural
This study	**NS-RH**	**701/319**	**0.683–0.689**[d]	**0.684–0.695**	**Yes**	**Yes**	**9 ML + RCS. Temporal val. EN recommended (0.684). Calibration + SHAP.**

[a]Zuo 2024 reports sensitivity/specificity on validation but not AUC separately. [b]Tian 2023 logistic regression AUC; their Random Forest achieved 0.996 on development (clear overfitting). [c]Bootstrap-corrected, no external cohort. [d]Range across top models (EN to SVM). Dev AUC is 10-fold CV (honest estimate), not resubstitution.

Key observations from the literature:

1. **Overfitting is pervasive.** Studies reporting dev AUC >0.85 consistently show large drops on validation: Shi 2025 (0.997→0.676), Tian 2023 RF (0.996→0.856), Porche 2023 (0.753→0.663). Our study avoids this by reporting CV-based development AUC (0.683–0.689) which matches validation AUC (0.684–0.695), with no train-val gap.
2. **External validation is rare.** Only 4 of the 14 comparison studies in Table 7 have external or temporal validation (Zuo 2024, Shi 2025, Porche 2023, Wei 2021). When validated externally, AUCs consistently fall in the 0.65–0.75 range, consistent with our result.
3. **Gynecologic POUR prediction models are scarce.** Only Zuo 2024 and Meng 2024 built prediction models for radical hysterectomy. Zuo's high dev AUC (0.897) includes *postoperative* predictors (UTI, pain score), which are unavailable at the time of preoperative risk counseling. Our model uses only preoperative/intraoperative variables.

4. **vs. Nantasupha 2018 (predecessor):** Same institution and senior author. They identified risk factors but reported no AUC or prediction model. We extend this with ML modeling and temporal validation. Our top predictor (vaginal cuff length) aligns with their surgical radicality finding. Their inter-surgeon variation (21–72%) represents unmeasured surgeon-level variation that our model cannot capture (see Limitations).
5. **vs. Shi 2025 (most comparable ML study):** Both use multiple ML algorithms for gynecologic POUR. Their GBDT achieved training AUC 0.997 but dropped to 0.676–0.710 on external validation — worse than our Elastic Net (0.684) which shows no train-val gap. Their approach exemplifies the overfit-then-collapse pattern we deliberately avoid.
6. **vs. Porche 2023 (prospective ML validation):** Their sensitivity collapsed from 68% to 23% on prospective validation. Our Elastic Net maintains sensitivity of 0.577 with specificity 0.697 — a more balanced and stable profile.

5 Addressing Research Questions

5.1 Primary: ML Model Accuracy for POD7 Voiding Prediction

Answer: Models clustered between AUC 0.63–0.70 on temporal validation. Stacking achieved the highest AUC (0.695) but is unsuitable for clinical use due to extreme calibration failure (slope 10.7). No pairwise comparison reached significance after Bonferroni correction. The **Elastic Net is recommended** for clinical use: calibration slope point estimate closest to 1.0 among full models (25-variable slope = 0.978, 95% CI 0.67–1.37; AUC 0.684), though the wide CI overlaps with LASSO and SVM, and the **5-variable version** (AUC 0.690, slope 1.075, 95% CI 0.71–1.51) is preferred for deployment due to simplicity and equivalent performance.

All models showed moderate discrimination. The AUC ceiling around 0.70 likely reflects inherent limitations in preoperative prediction — intraoperative factors (e.g., surgeon technique, nerve preservation quality) are known to be major determinants but are not measurable preoperatively.

5.2 Secondary 1: Factors Associated with Voiding Resumption

Answer: The consensus top predictors across all ML methods are:

1. **Average vaginal cuff length** — unanimously ranked #1. Longer vaginal cuff resection (proxy for extent of paravaginal dissection near pelvic nerves) is the strongest predictor. This aligns with the known mechanism: pelvic autonomic nerve injury during parametrial/paravaginal dissection.
2. **BMI** — higher BMI increases dysfunction risk, possibly through more difficult dissection or postoperative mobilization.
3. **Radicality class** — more radical procedures predict worse voiding outcomes.
4. **Preoperative lesion size (wardsize) / Age** — larger preoperative tumors may necessitate wider resection margins, indirectly increasing nerve injury risk. Older age is consistent with slower peripheral nerve recovery.

5.3 Secondary 2: Timing of Bladder Recovery

Answer: Partially addressable. The raw dataset contains a `delayPOD` variable for patients with voiding dysfunction at POD7, coded as:

- **1** = Recovery within 14 days postoperatively
- **2** = Recovery beyond 14 days postoperatively

Among 229 dysfunction patients with available timing data (out of 475 in the full cohort of 1,087 patients, or 440 in the 1,020 complete cases used for modeling; 48% with timing data):

- **60.7% (139/229)** recovered within 14 days postop
- **39.3% (90/229)** required more than 14 days for recovery

This indicates that the majority of patients with voiding dysfunction at POD7 recover relatively quickly (within the first 2 weeks), but a substantial minority (∼40%) have prolonged dysfunction. Precise average timing cannot be computed (data is binary, not continuous), and 52% of dysfunction patients have missing timing data, which may introduce selection bias in this secondary analysis.

6 Clinical Implications

Although the AUC of ∼0.70 represents moderate discrimination, the clinical value of this work extends beyond the headline metric:

Risk Stratification at the Point of Care

For clinical deployment, we recommend the **5-variable Elastic Net model** (age, BMI, parity, preoperative lesion size, and average vaginal cuff length) rather than the full 25-variable model. It achieves equivalent discrimination (AUC 0.690 vs. 0.684), good calibration (slope 1.075, 95% CI 0.71–1.51), and requires only variables routinely available during or immediately after surgery. We prefer the 5-variable Elastic Net over the 5-variable RCS Logistic (AUC 0.691 vs. 0.690, negligible difference) despite RCS having slightly better calibration (CITL −0.214 vs. −0.309; slope 0.945 vs. 1.075), because Elastic Net's linear form is simpler to implement in a web calculator and easier to interpret without spline basis functions. Note that average vaginal cuff length is measured intraoperatively, so the calculator is suitable for **early postoperative** risk estimation (e.g., at the end of surgery or POD1) rather than purely preoperative counseling. A clinician inputs these 5 variables to obtain a predicted probability of voiding dysfunction at POD7. This supports:

- **Patient counseling:** "Based on your profile, you have approximately a 35% chance of needing a catheter beyond day 7" — enabling informed consent and expectation management
- **Resource planning:** High-risk patients can be scheduled for earlier follow-up, and home catheterization training can be arranged preoperatively
- **Surgical planning:** Awareness that vaginal cuff length is the dominant predictor may encourage nerve-sparing technique refinement, particularly for patients already at elevated baseline risk

A prototype web-based calculator (Figure 7) demonstrates the deployment concept. With validated coefficients, this could be integrated into preoperative assessment workflows.

SHAP-Derived Insights for Surgical Practice

The SHAP analysis provides clinically actionable insights beyond traditional risk factor studies:

- **Vaginal cuff length** is the single most important predictor, unanimously ranked #1 across all methods. Unlike traditional logistic regression, SHAP quantifies the *magnitude* of its contribution for each individual patient — enabling patient-specific risk explanations rather than population-level odds ratios.
- **Surgical radicality** (class, parametrial length) dominates the feature rankings, consistent with the known mechanism of pelvic autonomic nerve injury. This reinforces the principle that nerve-sparing technique directly impacts voiding outcomes.
- **Non-modifiable factors** (age, BMI) identify patients who may warrant closer postoperative monitoring regardless of surgical approach.

Limitations of the Model for Clinical Use

An AUC of 0.70 means the model will misclassify a substantial proportion of patients. It should be used as **one input among many** in clinical decision-making — not as a standalone decision tool. The dominant unmeasured confounder (individual surgeon technique) accounts for much of the residual unpredictability. Nantasupha 2018 reported inter-surgeon dysfunction rates of 21–72%, variance that no preoperative model can capture.

7 Key Takeaways

1. **Modest but real predictive signal.** All ML models achieve AUC 0.63–0.70. This is consistent with the only other ML study for POUR (Porche 2023, AUC 0.663) and reflects the fundamental challenge: major determinants (surgeon technique, nerve preservation quality) are not capturable as preoperative predictors.
2. **No ML model significantly outperforms penalized logistic regression.** No AUC comparison vs. the pre-specified reference (Elastic Net) reached significance after Bonferroni correction for 8 tests

(two-sided bootstrap, $B = 2000$). This is expected with sample size $n \approx 1000$ and moderate signal strength.

3. **Elastic Net is the recommended model.** The 25-variable Elastic Net has the calibration slope point estimate closest to 1.0 among full models (slope 0.978, 95% CI 0.67–1.37; AUC 0.684), though CI overlap with LASSO and SVM precludes definitive ranking. For clinical deployment, the **5-variable Elastic Net** (AUC 0.690, slope 1.075, 95% CI 0.71–1.51) is preferred for its simplicity and equivalent performance.
4. **Vaginal cuff length dominates prediction.** Ranked #1 by all four importance methods. This is a measurable surgical parameter that could guide intraoperative decisions and postoperative management planning.
5. **Feature reduction from 25 to 14 predictors maintains performance.** Stage 2 models show equal or improved AUC with 14 features, suggesting the remaining 11 features contribute noise.
6. **Temporal drift is real.** Event rate dropped from 46.9% to 34.8% across eras. All models show negative CITL on the validation cohort. Recalibration-in-the-large corrects this effectively for well-calibrated models.
7. **This is the first ML prediction model specifically for POUR after nerve-sparing radical hysterectomy** (Shi 2025 applied ML to broader gynecologic abdominal-pelvic surgery). Extends the predecessor study (Nantasupha & Charoenkwan 2018) from the same institution with modern ML methodology and formal temporal validation.

8 Limitations

- Single-center retrospective data; requires external validation at other institutions.
- Complete-case analysis (93.8% retained). The 67 excluded cases had a higher dysfunction rate (52% vs. 43%) and were slightly from earlier years. Missingness was concentrated in two low-importance features (tumor appearance 3.2%, preoperative UTI 2.5%); no high-importance features (vaginal cuff length, BMI, age, preoperative lesion size, parity) had missing values. The exclusion pattern is not obviously inconsistent with MAR with respect to key predictors, but MCAR was not formally tested and the higher dysfunction rate among excluded cases suggests potential selection bias.
- Surgeon identity not included as a predictor due to confounding with era (some surgeons only active in certain years), but prior work shows it is a major determinant.
- Recovery timing data (`delayPOD`) available only as binary ($\leq$14 vs. >14 days) for 48% of dysfunction patients (229/475 with data; 246 missing); patients with missing timing data may differ systematically, so the 60.7%/39.3% split should be interpreted cautiously.
- AUC ceiling $\sim$0.70 suggests that preoperative predictors alone are insufficient; intraoperative or early postoperative data (e.g., day 1–3 PVR) would likely improve prediction.
- Feature selection performed on full development set before CV refit — minor information leakage in Stage 2 development estimates (validation estimates unaffected).
- Calibration slope 95% CIs are wide (e.g., Elastic Net 0.67–1.37) due to the limited validation sample ($n = 319$), reflecting non-trivial uncertainty around the point estimates.
- For some models (e.g., RCS Logistic), the validation AUC (0.691) marginally exceeds the development CV AUC (0.685). This is attributable to sampling noise given the wide confidence intervals (0.627–0.752) and does not indicate a methodological error.

9 Figures

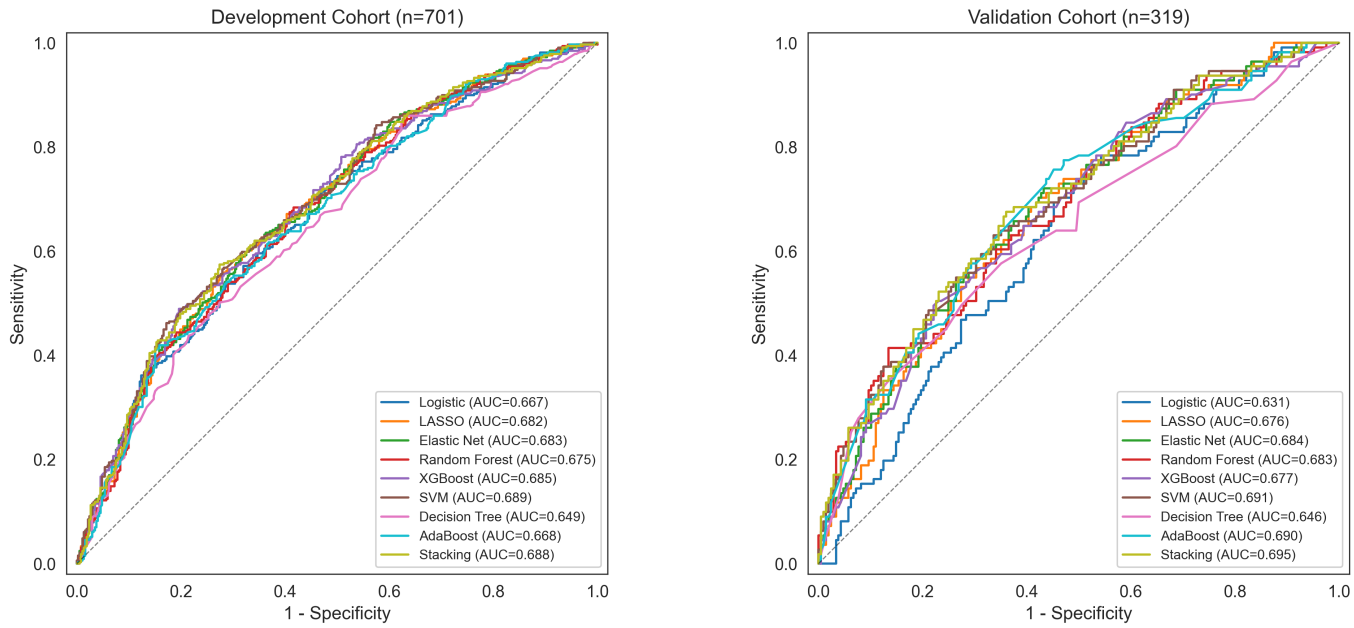

Figure 1: ROC curves for all models. Left: development cohort (10-fold CV). Right: temporal validation cohort.

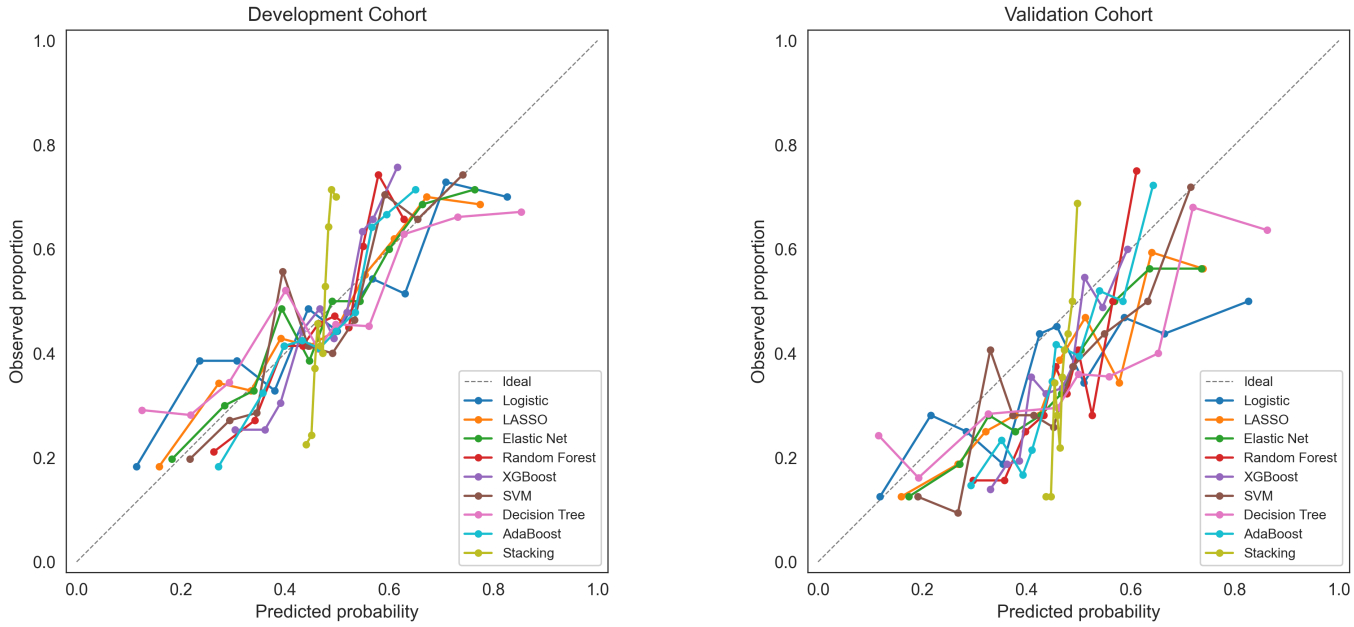

Figure 2: Calibration plots (quantile-binned deciles). Left: development. Right: validation. Elastic Net tracks closest to the ideal diagonal.

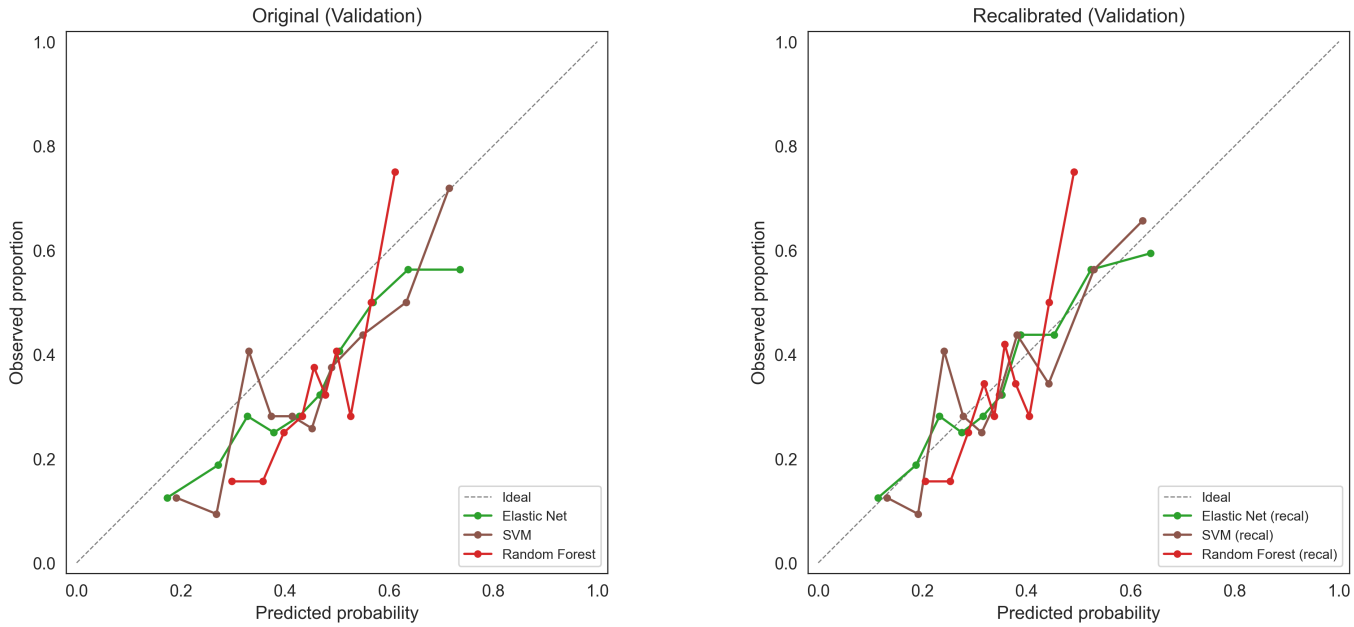

Figure 3: Calibration before (left) vs. after (right) recalibration-in-the-large on the validation cohort.

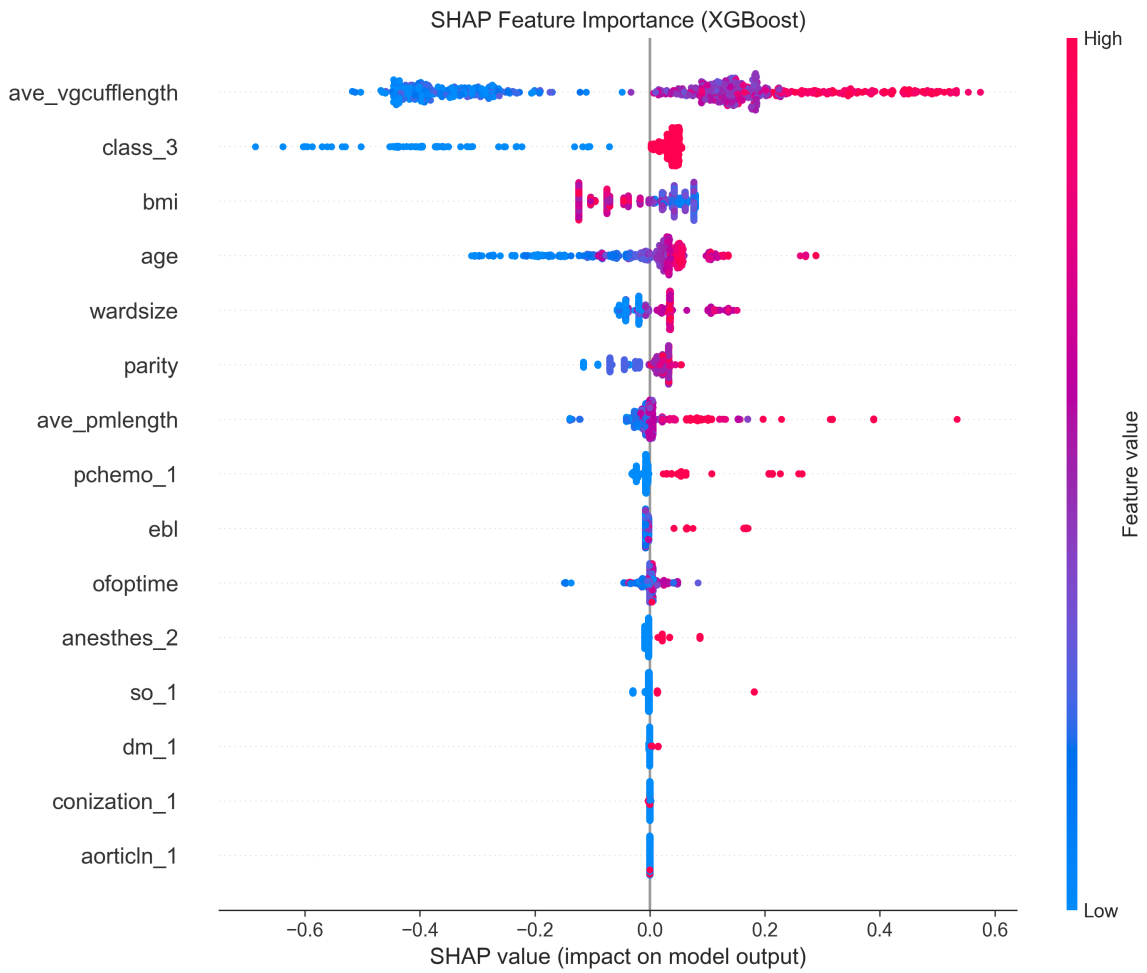

Figure 4: SHAP summary plot (XGBoost). Average vaginal cuff length dominates feature importance.

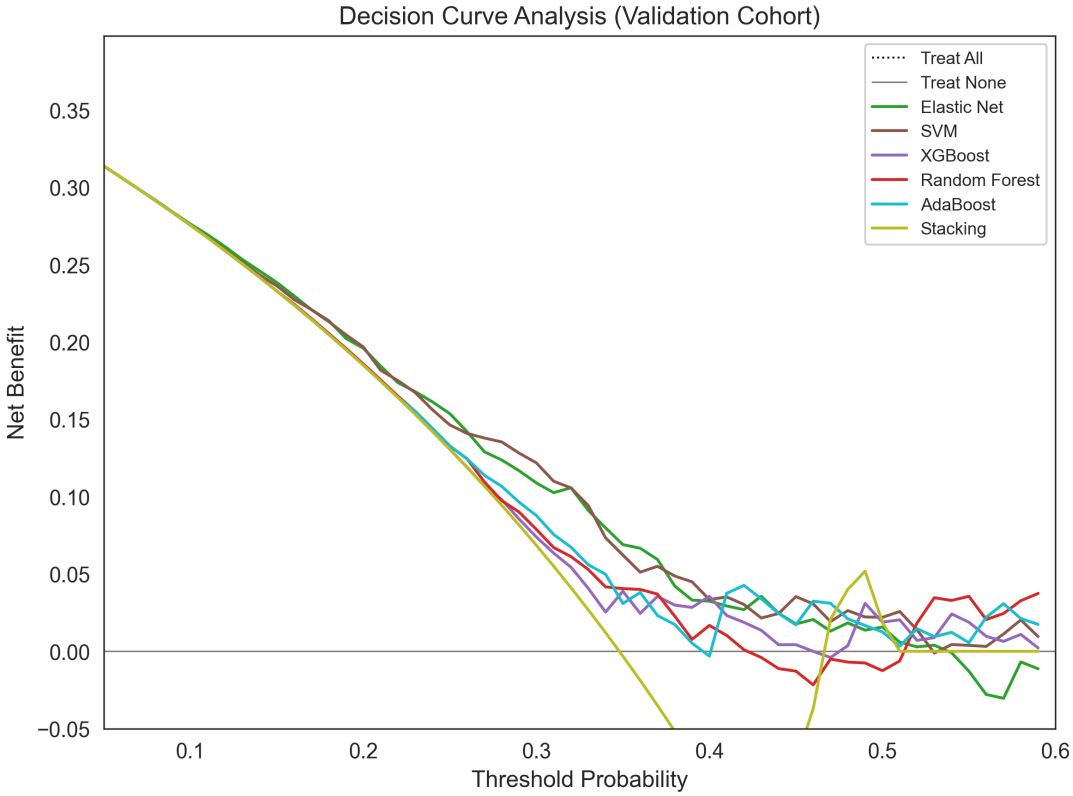

Figure 5: Decision curve analysis on the validation cohort. Most models provide net benefit above the "Treat All" strategy for threshold probabilities approximately 10–45% (the clinically plausible range where a preoperative intervention — e.g., catheter training, scheduled follow-up — would be considered); curves converge and become noisy beyond ∼50% due to sparse data at high thresholds.

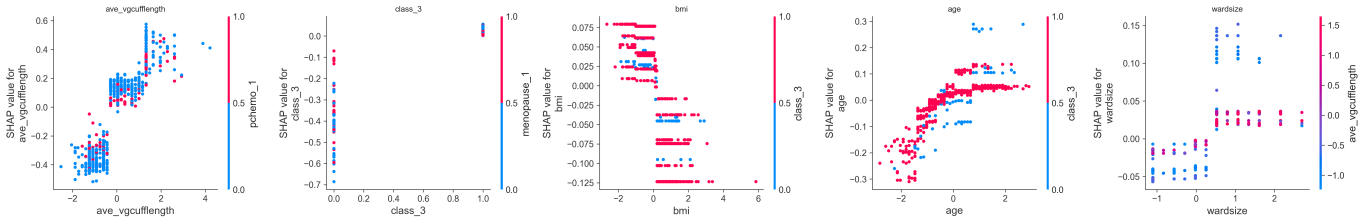

Figure 6: SHAP dependence plots for top 5 features showing the functional form of predictor-outcome relationships (predominantly linear or weakly non-linear).

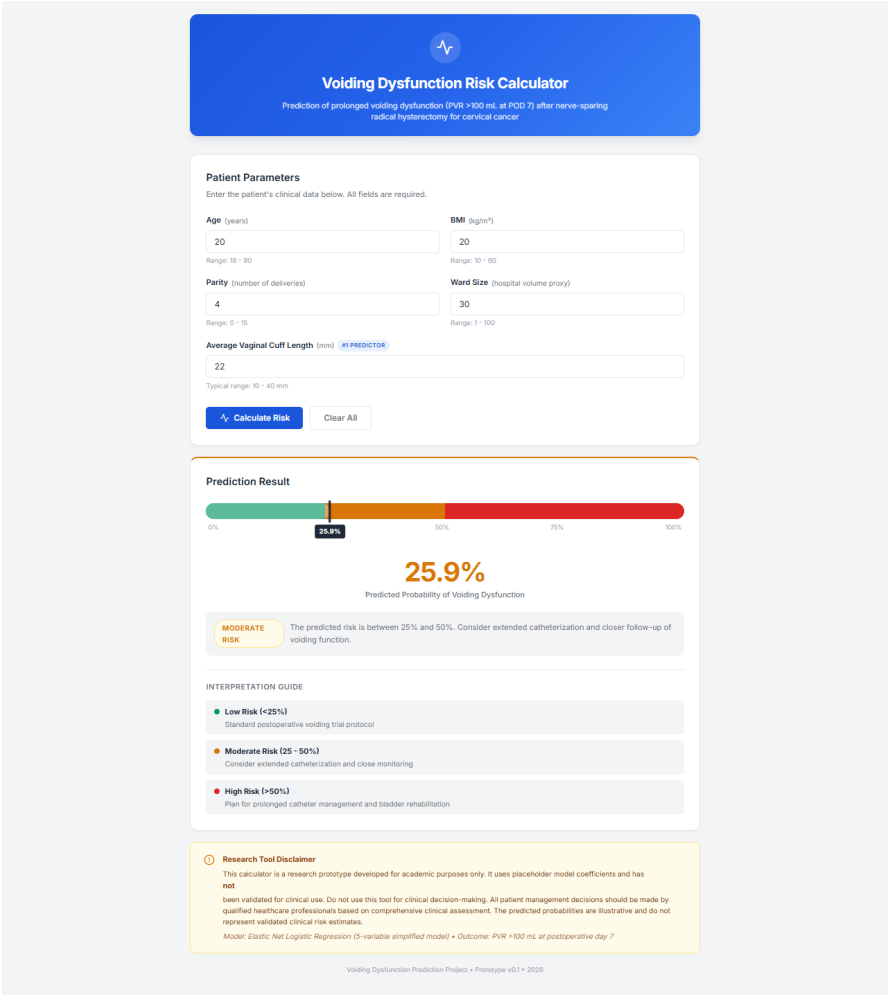

Figure 7: Prototype web-based risk calculator for voiding dysfunction. Uses the 5-variable Elastic Net model (age, BMI, parity, preoperative lesion size, average vaginal cuff length). Example shows a patient with moderate risk (25.9%). Coefficients shown are illustrative placeholders for demonstration purposes only; final deployment would use the validated model coefficients from the fitted Elastic Net. Available as a standalone HTML application (`app/index.html`).

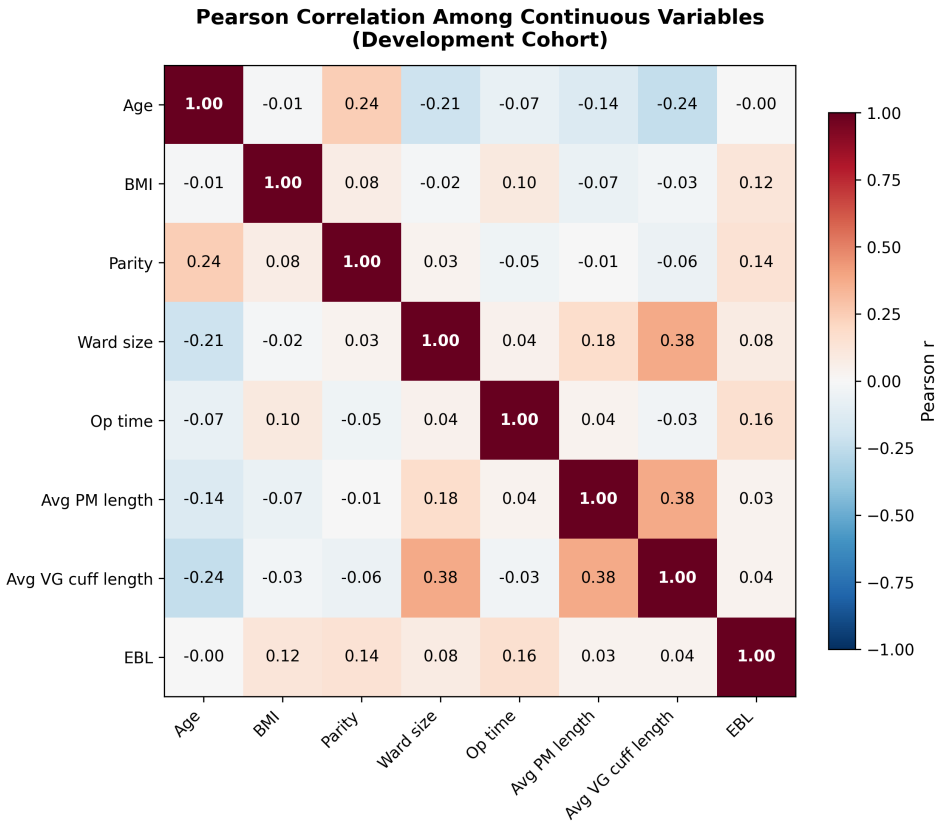

Figure 8: Pearson correlation matrix among continuous predictors (development cohort). Strongest correlations: avg vaginal cuff length – preoperative lesion size ($r = 0.38$), avg parametrial length – avg vaginal cuff length ($r = 0.38$), age – parity ($r = 0.24$). No multicollinearity concerns (all $|r| < 0.40$).

A Data Processing Pipeline

This appendix documents the complete data processing steps from raw Excel file to analysis-ready matrices.

Step 1: Load Raw Data

- **Source file:** `Voiding Dysfunction_03122025.xlsx`
- **Sheet:** `data_final_for anlysis_03122025` (note: original spelling preserved)
- **Raw dimensions:** 1,087 rows × 38 columns

Step 2: Recode and Derive Variables

1. **Outcome recoding:** The original `pod7voiding` variable is coded as 0 = dysfunction, 1 = normal voiding. We swap this ($0 \leftrightarrow 1$) so that **1 = dysfunction (event)** and 0 = normal voiding, following standard convention where 1 = the event of interest.
2. **Anesthesia recoding:** `anesthes` level 3 (combined/other, n small) merged into level 1 (general anesthesia), as done in the original R script.
3. **Derived continuous variables:**
   - $\text{BMI} = \text{wt}/(\text{ht}/100)^2$
   - $\text{ave_pmlength} = (\text{pmltlength} + \text{pmrtlength})/2$ (average parametrial length)
   - $\text{ave_vgcufflength} = (\text{antvg} + \text{postvg} + \text{ltvg} + \text{rtvg})/4$ (average vaginal cuff length)

   These replace the raw component measurements with clinically meaningful averages.

Step 3: Select Candidate Predictors

26 candidate predictors selected (matching the R analysis):

- **Continuous (8):** age, bmi, parity, wardsize, ofoptime (operative time), ave_pmlength, ave_vgcufflength, ebl (estimated blood loss)

- **Categorical (18):** disease, dm, menopause, hiv, previous_cs, preop_uti, conization, pchemo (prior chemotherapy), stage, incision, class (radicality class), anesthes, appendec, so (salpingo-oophorectomy), pelvicln, aorticln, finalhisto, appearance

  Plus the `year` column (for temporal splitting, not used as a predictor) and `pod7voiding` (outcome).

Step 4: Collapse Rare Categories

To prevent model instability from very small category levels:

Variable	Action	Rationale
`class`	Level 6 → 2	Level 6 has only $n = 2$; both are non-standard RH classes
`pelvicln`	Column dropped	Level 0 has $n = 11$ (1%); near-constant variable
`aorticln`	Levels 1,2 → 1	Both represent "yes" variants ($n = 17$ and $n = 27$)
`finalhisto`	Level 5 → 4	Level 5 ($n = 14$) is a rare histology subtype; merged into "other"
`appearance`	Level 3 → 4	Level 3 ($n = 11$) is rare; merged into "other/mixed"
`incision`	Level 4 → 3	Level 4 ($n = 20$) is a rare transverse variant

After dropping `pelvicln`, the final predictor count is **25** (8 continuous + 17 categorical).

Step 5: Complete-Case Filter

- Rows with any missing values dropped: 1,087 → 1,020 (93.8% retained)
- 67 rows excluded (6.2% missingness, scattered across variables)

Step 6: Temporal Split

Splitting by the `year` column (Buddhist calendar year):

- **Development cohort:** Year 2548–2557 (= 2005–2014 CE), $n = 701$, events = 329 (46.9%)
- **Validation cohort:** Year 2558–2566 (= 2015–2023 CE), $n = 319$, events = 111 (34.8%)

This is a strict temporal split — no data leakage between cohorts. The event rate drop (46.9% → 34.8%) represents real temporal drift, likely reflecting improved nerve-sparing technique over time.

Step 7: Prepare Feature Matrices

1. Separate outcome (y) from predictors (X): $y =$ `pod7voiding`, $X =$ remaining 25 columns
2. Cast all categorical columns to `category` dtype
3. Preprocessing pipeline (applied inside each CV fold to prevent leakage):
   - **Continuous features:** StandardScaler (zero mean, unit variance)
   - **Categorical features:** OneHotEncoder (drop if binary; group infrequent categories with $n < 5$)
4. Final dimensions: X_{dev}: $(701, 25)$, X_{val}: $(319, 25)$

Step 8: Cross-Validation Strategy

- **Method:** 10-fold stratified cross-validation on the development cohort
- **Stratification:** Preserves the outcome class ratio (∼47% events) in each fold
- **Hyperparameter tuning:** GridSearchCV nested inside CV, optimizing AUC
- **Threshold selection:** Youden's J statistic (max(sensitivity + specificity − 1)) derived from development cohort CV predictions, then applied to the held-out validation cohort
- **Random seed:** 42 (for reproducibility)

Summary Diagram

```
Excel (1,087 x 38)
  |-- Sheet: "data_final_for anlysis_03122025"
  |
  v
Recode outcome (0<->1) + derive BMI, ave_pmlength, ave_vgcufflength
  |
  v
Select 26 predictors + year + outcome
  |
  v
Collapse rare categories --> 25 predictors (pelvicln dropped)
  |
  v
Complete cases: 1,087 --> 1,020 (93.8%)
  |
  v
Temporal split by year
  |-- Dev (2005-2014): n=701, 329 events (46.9%)
  |-- Val (2015-2023): n=319, 111 events (34.8%)
  |
  v
StandardScaler (continuous) + OneHotEncoder (categorical)
  |
  v
X_dev (701,25), y_dev (701,)  |  X_val (319,25), y_val (319,)
  |
  v
10-fold stratified CV --> GridSearchCV --> Youden threshold
```

B Hyperparameter Search Spaces and Best Values

All models were tuned via GridSearchCV (10-fold stratified CV, scoring = AUC). Table 8 shows the full search grids; Table 9 shows the selected values.

Table 8: Hyperparameter search grids for each model.

Model	Grid Searched
Logistic	No tuning (unpenalized, solver = saga)
LASSO	C: {0.001, 0.01, 0.1, 0.5, 1, 5, 10}
Elastic Net	C: {0.01, 0.1, 0.5, 1, 5, 10}; l1_ratio: {0.1, 0.3, 0.5, 0.7, 0.9}
Random Forest	n_estimators: {200, 500}; max_depth: {3, 5, 8, None}; min_samples_leaf: {5, 10, 20}
XGBoost	n_estimators: {100, 300, 500}; max_depth: {3, 5, 7}; learning_rate: {0.01, 0.05, 0.1}; subsample: {0.8, 1.0}; colsample_bytree: {0.8, 1.0}
SVM (RBF)	C: {0.1, 1, 10, 50}; gamma: {scale, 0.01, 0.001}
Decision Tree	max_depth: {3, 5, 8, 10, None}; min_samples_leaf: {5, 10, 20, 30}; min_samples_split: {10, 20, 50}
AdaBoost	n_estimators: {50, 100, 200, 500}; learning_rate: {0.01, 0.05, 0.1, 0.5, 1.0}
Stacking	Base learners use Stage 1 tuned hyperparameters (fixed, not re-optimized within stacking CV): EN (C=0.1, l1=0.1), RF (200, depth=5), XGB (100, depth=3, lr=0.01), SVM (C=1, gamma=0.01), AdaBoost (50, lr=0.5). Only meta-learner C is tuned: {0.01, 0.1, 1, 10}

Table 9: Best hyperparameters selected by GridSearchCV (highest mean CV AUC).

Model	**Best Parameters**
Logistic	None (unpenalized)
LASSO	C = 0.5
Elastic Net	C = 0.1, l1_ratio = 0.1 (nearly pure L2)
Random Forest	n_estimators = 200, max_depth = 5, min_samples_leaf = 5
XGBoost	n_estimators = 100, max_depth = 3, learning_rate = 0.01, subsample = 1.0, colsample = 1.0
SVM	C = 1, gamma = 0.01
Decision Tree	max_depth = 8, min_samples_leaf = 30, min_samples_split = 10
AdaBoost	n_estimators = 50, learning_rate = 0.5
Stacking	meta-learner C = 0.01

Notable patterns: Elastic Net selected C = 0.1 with l1_ratio = 0.1, indicating strong regularization with nearly pure L2 (ridge) penalty — consistent with the finding that most features contribute small, diffuse effects rather than sparse strong signals. XGBoost selected conservative hyperparameters (max_depth = 3, n_estimators = 100, learning_rate = 0.01), reflecting the limited signal available.